

ARIA-NSF: A Differentiable Klatt-Style Neural Source-Filter Synthesizer for Small-Data Phonetic Stimulus Generation

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Abstract

Speech-science experiments often require synthetic stimuli that are natural enough for repeated listening, but also explicitly controllable along phonetically meaningful dimensions such as fundamental frequency, formant frequency, spectral tilt, intensity, and voice quality. Modern neural vocoders provide high perceptual quality, but their data requirements and latent acoustic representations can make them difficult to use in small-data phonetic studies. Classical formant synthesis provides interpretability and control, but often lacks the naturalness required for ecological experimental stimuli. This paper presents ARIA-NSF, a differentiable Klatt-style neural source-filter synthesizer designed for small-data, single-speaker phonetic stimulus generation. The model combines an analytic glottal excitation controlled by F_0 and R_d , a learned noise component, and a vocal-tract cascade that combines deterministic formant/all-pole sections with learned biquad residual sections. In the current configuration, the vocal-tract cascade contains three deterministic biquad sections and eight learned biquads, yielding an LPC-equivalent order of $2(3 + 8) = 22$. Rather than proposing a general-purpose neural vocoder, ARIA-NSF targets a narrower experimental setting: controllable copy synthesis and acoustic manipulation from laboratory-scale speech recordings. We describe the architecture, the pilot Mandarin corpus, and a concrete evaluation plan covering acoustic error, parameter disentanglement, baseline comparison, and perceptual validation.

Keywords: neural source-filter model; formant synthesis; differentiable DSP; phonetic stimulus generation; small-data speech synthesis; voice quality

1. Introduction

Controlled speech synthesis occupies an unusual position between speech technology and phonetic science. In text-to-speech and neural vocoding, the primary goal is often perceptual naturalness, speaker similarity, or large-scale generalization. In phonetic and psycholinguistic experiments, however, the research goal is frequently different: the stimulus must be natural enough to be listened to repeatedly, but the experimental manipulation must also be interpretable, precise, and preferably low-dimensional. A phonetician may need to shift F_0 to create a tone or intonation continuum, move F_1 and F_2 independently to construct a vowel continuum, or alter source-related parameters to probe breathy, creaky, or pressed voice quality. A system that produces plausible audio but hides these variables inside a high-dimensional latent representation is therefore not always suitable for experimental work.

This requirement is especially challenging in typical speech-science settings. Unlike modern speech-generation pipelines, which may rely on many hours or thousands of hours of training data, experimental materials are often collected for a specific study. A dataset may contain only 5–60 minutes of speech from one speaker, recorded under a controlled design for a particular phenomenon such as tone sandhi, categorical perception, prosodic prominence, vowel normalization, or phonation type. Such datasets are valuable precisely because they are tailored to a hypothesis, but they are usually too small to support large generic neural vocoders without overfitting, artifacts, or loss of controllability.

Classical source-filter theory (Fant 1960) and formant synthesis (Klatt 1980) offer the right conceptual primitives: a source, a vocal-tract filter, and interpretable acoustic controls. Tools such as Praat (Boersma 2001) remain central in phonetics because they support analysis and manipulation in terms that phoneticians understand.

Yet purely classical synthesis can sound unnatural, especially when used to create full experimental stimuli rather than isolated demonstrations. Conversely, modern neural vocoders and neural source-filter models can generate high-quality waveforms (Wang et al. 2020), but their control variables are often less directly aligned with experimental manipulations.

This paper presents ARIA-NSF, a small-data differentiable Klatt-style neural source-filter synthesizer. The central design hypothesis is that phonetic stimulus generation benefits from a hybrid allocation of responsibility: parameters that are experimentally meaningful and acoustically well-understood should remain analytic or deterministic, while the residual aspects of timbre and spectral detail can be learned from data. In the current implementation, F_0 , F_1 , F_2 , bandwidths, gain, spectral tilt, and glottal-source parameters are exposed as controllable variables. A small neural model predicts or refines the remaining control streams needed for copy synthesis.

The intended contribution is not a universal TTS system. Instead, ARIA-NSF targets a narrower but common research use case: creating high-quality, interpretable speech stimuli from a small, single-speaker corpus. The paper makes three contributions:

1. It formulates the requirements of neural speech synthesis for phonetic stimulus generation as a small-data, high-control source-filter problem.
2. It introduces a differentiable Klatt-style architecture that combines an analytic glottal source, shaped noise, deterministic formant sections, and learned residual biquads.
3. It defines a pilot Mandarin evaluation protocol for measuring copy-synthesis quality, manipulation reliability, and unwanted acoustic coupling.

2. Related Work

2.1 Classical Formant Synthesis and Phonetic Manipulation

The source-filter model of speech production provides a compact account of speech as the interaction between a laryngeal source and a vocal-tract filter (Fant 1960). Klatt’s cascade/parallel formant synthesizer (Klatt 1980) made this account operational by exposing parameters such as formant frequencies, bandwidths, voicing amplitude, aspiration, frication, and spectral tilt. For phonetic experiments, this level of control remains attractive because it allows a researcher to manipulate one acoustic dimension while holding others approximately constant.

The limitation is perceptual quality. Even when a classical formant synthesizer accurately implements an acoustic hypothesis, the resulting stimulus may sound synthetic in ways that affect listener behavior. This matters because perceptual experiments are not only tests of acoustic discriminability; they are also embedded in listeners’ expectations about natural speech. A stimulus continuum with audible artifacts may therefore confound the intended manipulation.

Praat (Boersma 2001) and LPC-based copy synthesis remain widely used because they allow practical analysis and manipulation of pitch and formant-related properties. However, inverse filtering and source-filter separation are imperfect. If formant information remains in the residual excitation, formant manipulation can introduce artifacts or unintended acoustic coupling.

2.2 Neural Source-Filter and Differentiable DSP Models

Neural source-filter models reintroduced speech-production structure into neural waveform generation. Wang et al. (2020) proposed an NSF framework in which a source module generates an excitation signal, a neural filter module transforms it into a waveform, and a condition module preprocesses acoustic features. This design preserves explicit F_0 -driven excitation while avoiding autoregressive generation.

Differentiable digital signal processing (DDSP) similarly argues that neural audio models can benefit from embedding known signal-processing operators inside trainable systems (Engel et al. 2020). Rather than learning the entire waveform generator as an unconstrained network, DDSP systems can learn control parameters for interpretable modules such as oscillators, filters, and reverberation. This is especially relevant for small-data settings, where inductive bias can reduce the amount of data required to learn a usable synthesizer.

GOLF (Yu and Fazekas 2024) is a particularly close reference point. It uses glottal-flow wavetables as the harmonic source and LPC filters to model the vocal tract, achieving a compact and efficient singing voice synthesizer.

GOLF shows that voice-production constraints can be integrated into differentiable synthesis while maintaining competitive quality and efficiency. ARIA-NSF follows the same general philosophy but targets speech-science stimulus generation rather than singing voice synthesis, and gives priority to explicit manipulation of F_0 , F_1/F_2 , and source-shape parameters.

HiFi-Glot (Gu et al. 2026) addresses a related problem from the direction of high-fidelity neural formant synthesis. It combines neural excitation with differentiable resonant/all-pole filters to provide explicit formant control at high perceptual quality. Compared with such full-band, large-capacity systems, ARIA-NSF is deliberately smaller and more specialized. It sacrifices some generality in favor of small-data trainability, explicit experimental controls, and a structure that remains close to traditional formant synthesis.

2.3 Source-Filter Positioning

The source-filter taxonomy used in uSFGAN (Yoneyama et al. 2021) is useful for locating ARIA-NSF among conventional vocoders, neural source-filter models, and fully neural waveform generators. In that taxonomy, excitation generation and resonance filtering can each be implemented as deterministic/parametric modules, neural modules, or a unified neural waveform generator. Figure 1 adapts this view and places ARIA-NSF as a dual-hybrid source-filter model.

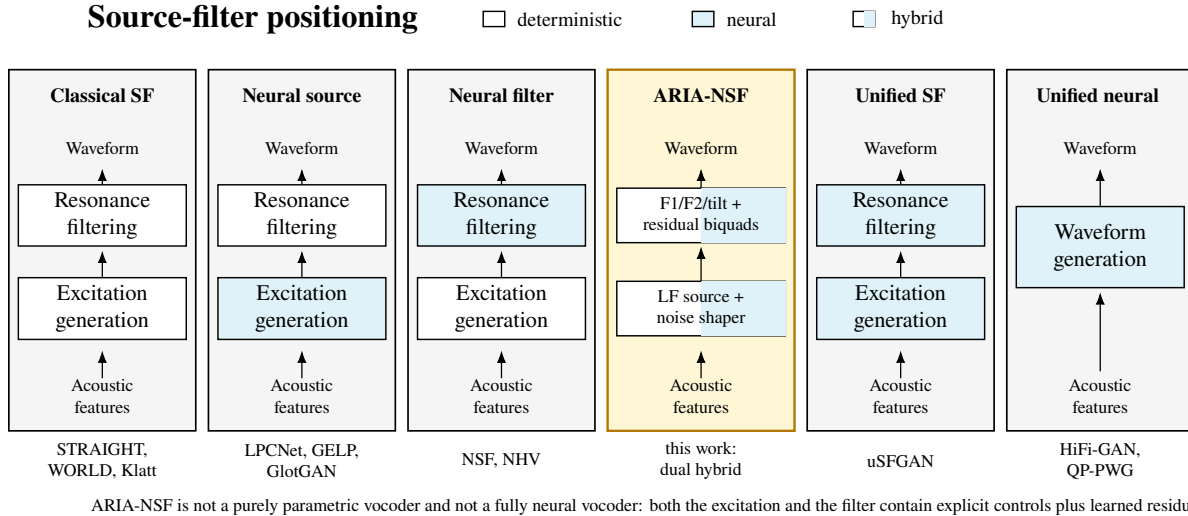


Figure 1: Positioning of ARIA-NSF within source-filter vocoder architectures. The diagram is adapted from the source-filter taxonomy in uSFGAN (Yoneyama et al. 2021), but adds the ARIA-NSF column to emphasize that both excitation generation and resonance filtering are hybrid rather than purely deterministic or purely neural.

ARIA-NSF is therefore better described as structurally hybrid than as a literal 50/50 split between deterministic and neural components. On the excitation side, the periodic glottal source is generated deterministically from F_0 and R_d , while the aperiodic/noise component is learned or conditioned. On the filter side, the experimentally important vocal-tract controls remain deterministic and interpretable, while learned residual biquads supply speaker-specific spectral detail. The model thus splits responsibility in both places: explicit DSP handles the variables that should remain editable, and neural residual capacity handles details that are difficult to specify by hand.

3. Design Requirements

ARIA-NSF is designed around five requirements that are common in phonetic stimulus generation but less central in general-purpose neural vocoding.

R1. Small-data trainability. The system should be usable with 5–60 minutes of single-speaker recordings, because phonetic corpora are often collected for a narrow experimental question rather than for large-scale speech

generation.

R2. Explicit intervention variables. Experimental variables such as F_0 , F_1/F_2 , bandwidth, gain, spectral tilt, and R_d should remain directly editable. They should not be recoverable only as post-hoc measurements from a generated waveform.

R3. Low cross-parameter coupling. Manipulating one dimension should minimally disturb others. For example, an F_1 manipulation should not substantially alter F_0 , energy, or source-shape cues.

R4. Residual expressivity. Purely analytic formant synthesis may not capture enough speaker-specific spectral detail for natural-sounding stimuli. The architecture therefore needs limited learned capacity, but this capacity must be constrained enough that it does not erase interpretability.

R5. Laboratory accessibility. Training should be feasible on consumer or modest institutional GPUs. This requirement is practical rather than theoretical, but it is important for speech-science laboratories without sustained access to large compute clusters.

These requirements motivate a dual-hybrid source-filter design: deterministic components implement the experimental variables, while learned residual components model aspects of both the excitation and the filter that are difficult to specify analytically.

4. Model

Figure 2 gives a schematic overview of the current ARIA-NSF pipeline. The architecture operates at two rates. Frame-rate acoustic controls are predicted, extracted, or specified by the user; these controls are then upsampled to the audio sample rate for differentiable waveform synthesis.

4.1 Sample-Rate Excitation

Given sampling rate f_s , the deterministic phase trajectory is obtained by integrating the frame- or sample-rate F_0 contour:

$$\phi_n = \phi_{n-1} + 2\pi \frac{F_{0,n}}{f_s}.$$

The voiced excitation is generated by an LF-style glottal source:

$$e_n = g_{\text{LF}}(\phi_n; R_{d,n}),$$

where R_d controls the glottal pulse shape and is intended to support continuous manipulation of phonation-related dimensions such as pressed or breathy voice. This formulation follows the spirit of LF-family source modeling (Fant et al. 1985), but is used here as a differentiable component inside a neural source-filter synthesizer rather than as a complete standalone voice model.

A noise branch models turbulent or aperiodic energy. White noise ϵ_n is transformed by a learned or conditioned noise shaper:

$$\eta_n = \mathcal{N}(\epsilon_n; m_n),$$

where m_n denotes the sample-rate noise control stream. In contrast to architectures that filter harmonic and noise components through separate output paths, ARIA-NSF currently combines excitation and shaped noise before the vocal-tract filter:

$$u_n = e_n + \eta_n.$$

This design choice follows the intended speech-production interpretation: the periodic and aperiodic components form a source-like excitation, which is then shaped by the vocal tract.

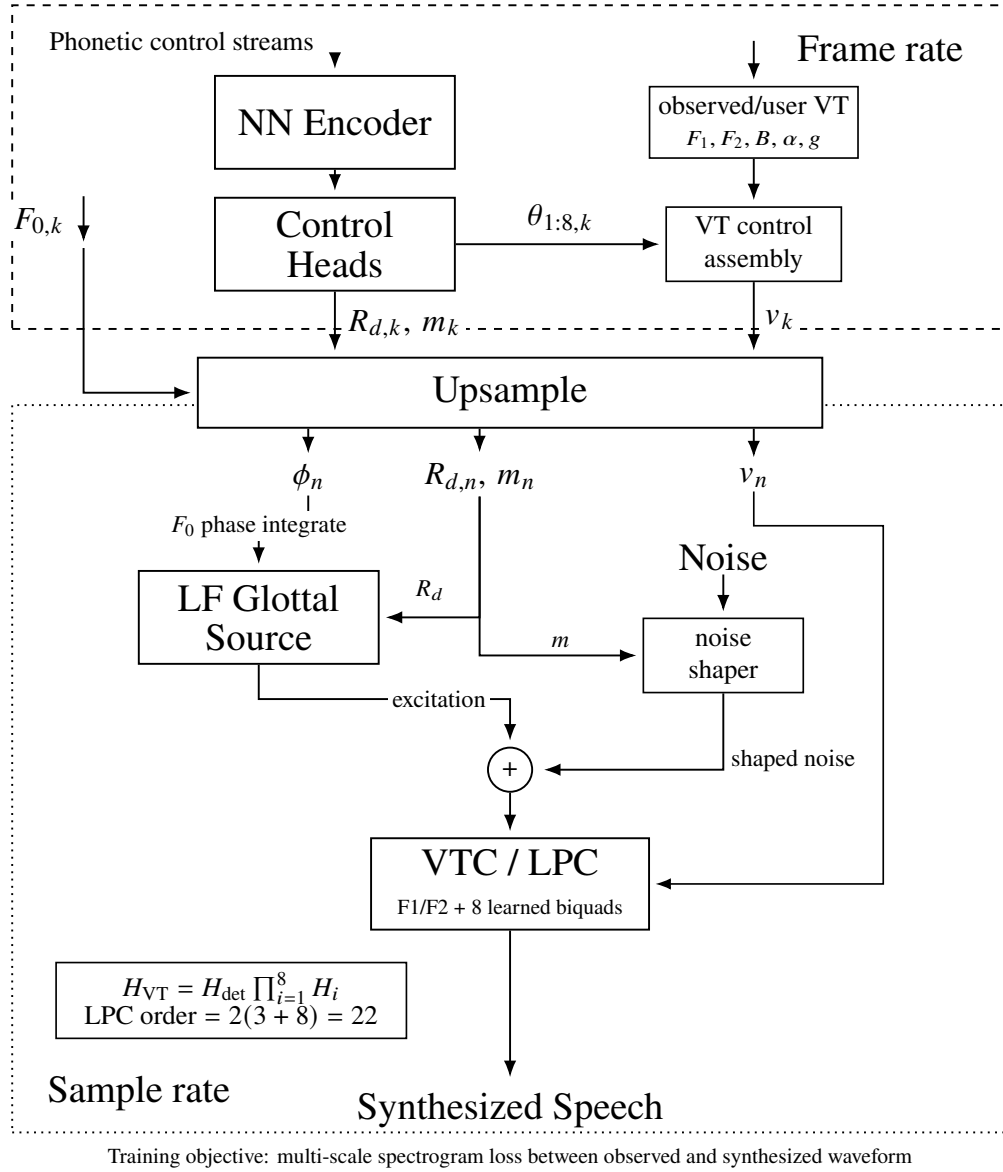


Figure 2: Current ARIA-NSF source-filter diagram. The glottal excitation and shaped noise are summed before passing through the vocal-tract cascade / LPC filter.

4.2 Vocal-Tract Cascade

The output waveform is generated by passing the combined excitation through a time-varying vocal-tract cascade:

$$y_n = H_{\text{VT}}(z; v_n)u_n.$$

The filter is factorized into deterministic, interpretable sections and learned residual sections:

$$H_{\text{VT}}(z; v_n) = H_{\text{det}}(z; F_{1,n}, F_{2,n}, B_{1,n}, B_{2,n}, \alpha_n, g_n) \cdot \prod_{i=1}^8 H_i(z; \theta_{i,n}).$$

Each biquad section can be written as an all-pole resonator:

$$H_i(z; \theta_i) = \frac{G_i}{1 + a_{i,1}z^{-1} + a_{i,2}z^{-2}}.$$

In the current configuration, the deterministic component contains three biquad sections and the neural residual component contains eight learned biquads. The resulting LPC-equivalent order is:

$$\text{order} = 2(3 + 8) = 22.$$

This configuration is intended to make the first formant-related degrees of freedom explicitly controllable while leaving enough residual capacity to learn speaker-specific spectral detail that would be difficult to hand-code.

4.3 Training Objective

The pilot implementation is trained as a copy-synthesis model using a multi-scale spectral reconstruction objective:

$$\mathcal{L}_{\text{MSSTFT}} = \sum_{r \in \mathcal{R}} (\| |S_r(x)| - |S_r(y)| \|_1 + \| \log(|S_r(x)| + \epsilon) - \log(|S_r(y)| + \epsilon) \|_1),$$

where x is the target waveform, y is the synthesized waveform, and S_r is an STFT operator at resolution r . This objective is consistent with the spectral-loss tradition in DDSP and neural source-filter work (Engel et al. 2020; Wang et al. 2020; Yu and Fazekas 2024), while keeping the training procedure lightweight.

5. Pilot Corpus and Experimental Conditions

The current pilot corpus consists of Mandarin monosyllabic word recordings collected for phonetic research. The recording session contains approximately 1,400 word tokens, with about 30 minutes of effective usable audio after trimming and quality control. This corpus is representative of a common phonetic-laboratory scenario: the material is carefully designed for a linguistic question, but it is much smaller than the datasets normally used to train high-capacity neural vocoders.

The pilot evaluation is organized around three operations:

1. **Copy synthesis:** reconstructing held-out utterances from extracted or predicted acoustic controls.
2. **Pitch manipulation:** modifying F_0 trajectories while preserving other source-filter properties.
3. **Vowel and source manipulation:** independently altering F_1/F_2 and R_d to create continua relevant to vowel perception and voice-quality perception.

The planned experimental conditions are summarized in Table 1. The 30-minute condition is treated as the primary pilot setting. The 5- and 10-minute conditions test whether the architecture remains useful when the available speech material is closer to the lower bound of many phonetic laboratory datasets.

Table 1. Planned pilot and ablation conditions.

Condition	Training material	Purpose
Full pilot	approximately 30 min	Main copy-synthesis and manipulation condition
Reduced data	approximately 10 min	Small-data robustness check
Minimal data	approximately 5 min	Lower-bound feasibility check
Ablation: no learned biquads	approximately 30 min	Tests contribution of residual spectral modeling
Ablation: deterministic formants disabled	approximately 30 min	Tests whether explicit formant control is necessary
Ablation: no shaped noise	approximately 30 min	Tests contribution of aperiodic excitation

Early listening during development suggests that the model can converge on the 30-minute corpus and can still produce usable output under 5–10 minute training conditions. These observations motivate the evaluation but should not be interpreted as final benchmark results.

The model is lightweight enough to train on a laptop-class RTX 4060 GPU. On an A100-class server GPU, training time is typically on the order of tens of minutes for the current single-speaker setting. This resource profile is important for speech-science laboratories, where continuous access to large-scale GPU clusters is often unrealistic.

6. Evaluation Protocol

A formal evaluation should measure both synthesis quality and manipulation validity. For phonetic stimulus generation, these two dimensions are equally important. A system that sounds natural but fails to control the intended variable is not useful for controlled experiments; likewise, a precisely controlled but unnatural stimulus may not support ecologically valid listening behavior.

6.1 Baselines and Ablations

The evaluation should compare ARIA-NSF against baselines that represent both phonetic practice and neural synthesis practice:

Table 2. Proposed baselines and ablation systems.

System	Role in evaluation
Natural recording	Upper reference for perceptual quality
Praat / LPC copy synthesis	Conventional phonetic manipulation baseline
DDSP harmonic-plus-noise	Differentiable synthesis baseline without explicit vocal-tract cascade
HiFi-Glot-style source-filter model	Strong neural formant-synthesis reference, where feasible
ARIA-NSF without learned biquads	Tests whether residual capacity improves quality
ARIA-NSF without deterministic formant sections	Tests whether explicit formant control is responsible for manipulation accuracy
ARIA-NSF without shaped noise	Tests whether aperiodic excitation improves naturalness

6.2 Acoustic Evaluation

For copy synthesis, standard reconstruction metrics can be reported, including multi-scale STFT loss, mel-cepstral distortion, F_0 error, and spectral-envelope error. For manipulation tasks, the more important metric is target-tracking error. For formants:

$$E_{F_i} = \frac{1}{T} \sum_t \left| \hat{F}_i(y_t) - F_i^{\text{target}}(t) \right|, \quad i \in \{1, 2\}.$$

The same logic can be applied to F_0 , intensity, spectral tilt, and any estimated correlate of voice quality. To measure unwanted coupling, one can compute a manipulation cross-effect matrix. For example, when F_1 is manipulated, the analysis should report not only the F_1 error but also the induced changes in F_0 , F_2 , energy, tilt, and R_d -related measures.

For reporting, the most important outcome is not a single aggregate score but a separation between quality, target accuracy, and side effects:

Dimension	Example metric	Interpretation
Copy quality	MS-STFT, MCD, spectral-envelope error	How well the model reconstructs held-out speech
Pitch control	F_0 RMSE after manipulation	Whether deterministic excitation tracks the target
Formant control	F_1/F_2 target-tracking error	Whether vowel-space manipulations are accurate
Source control	acoustic/perceptual correlate of R_d	Whether source-shape manipulation is interpretable
Coupling	cross-effect matrix	Whether manipulating one parameter disturbs others
Efficiency	training time, inference RTF, model size	Whether the system is usable in laboratory settings

6.3 Perceptual Evaluation

For copy synthesis, a MUSHRA-like or CMOS listening test can compare ARIA-NSF with natural recordings, Praat/LPC copy synthesis, and neural baselines such as DDSP harmonic-plus-noise or a HiFi-Glot-style source-filter system. For manipulation, the evaluation should be task-based rather than only naturalness-based. Suitable designs include:

1. identification or goodness-rating tests along an F_1/F_2 vowel continuum;
2. tone or intonation perception tests along an F_0 continuum;
3. voice-quality rating tests along an R_d continuum;
4. ABX or oddity tests to assess whether listeners perceive the intended dimension rather than artifacts.

The key criterion is whether the synthesized stimuli support the same phonetic contrast that the experiment is designed to test.

7. Discussion

ARIA-NSF is motivated by a gap between two mature traditions. Classical formant synthesis gives phoneticians transparent control, but it can sound artificial. Neural vocoders provide high-quality audio, but their control variables are often too entangled for experimental stimulus generation. A differentiable Klatt-style neural source-filter model offers a middle path: keep the experimentally meaningful variables explicit, and learn the residual structure needed for naturalness.

For speech scientists, the main advantage is experimental control. F_0 , F_1/F_2 , spectral tilt, intensity, and R_d can be treated as intervention variables rather than as post-hoc measurements of a generated waveform. For machine-learning researchers, the model is an example of structured inductive bias: by placing source-filter assumptions inside the architecture, the system can operate in a data regime where large black-box neural vocoders are poorly matched to the task.

The learned biquad cascade is both useful and potentially risky. It increases expressive power and improves speaker-specific spectral modeling, but it may also reintroduce coupling between parameters that are meant to remain independent. Future work should analyze the learned sections directly, for example by tracking pole trajectories, inspecting formant interactions, and ablating the learned residual cascade.

The current pilot implementation should therefore be read as a system and evaluation proposal rather than as a completed benchmark. Its strongest claim is architectural: phonetic stimulus generation may benefit from keeping experimentally meaningful parameters analytic while assigning only the residual spectral detail to learned modules. The empirical claim will require controlled comparison against Praat/LPC manipulation, DDSP-style harmonic-plus-noise synthesis, and a stronger neural source-filter or HiFi-Glot-inspired model.

8. Conclusion

This paper introduced ARIA-NSF, a differentiable Klatt-style neural source-filter synthesizer for small-data phonetic stimulus generation. The model combines deterministic source and formant controls with learned residual spectral modeling, aiming to preserve the interpretability of classical formant synthesis while improving the naturalness of copy synthesis and manipulation. The current architecture uses an LF-style glottal excitation, a shaped noise branch, and a vocal-tract cascade consisting of deterministic all-pole sections plus eight learned biquads. The revised evaluation plan focuses on the criteria that matter most for phonetic experiments: target accuracy, low cross-parameter coupling, perceptual adequacy, and small-data trainability. The next step is to complete the objective and perceptual evaluations needed to turn this system draft into a fully supported conference submission.

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Data Availability

The current pilot corpus contains speech recordings collected for phonetic research and is not yet publicly released. A demo page with selected copy-synthesis and manipulation examples is available at https://n1r.github.io/ARIA_nsf/.

Ethics Statement

Future perceptual evaluation will require informed consent from listeners and appropriate handling of speaker recordings. No perceptual listener data are reported in this draft.

Author Contributions

Yiran Ding designed the model, prepared the pilot corpus, implemented the current system, and wrote the draft.

Conflict of Interest

The author declares no conflict of interest.

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AI Assistance Disclosure

AI writing assistance was used to organize the draft structure and polish language. The author is responsible for all technical claims, experiments, and final manuscript content.

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